



Seismic Isolation Bearing Design

002 Compression Stiffness

Example 04 - rivtbook

proj. 0004



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0.2i | Shape Factor

COMPRESSION BLOCK¹

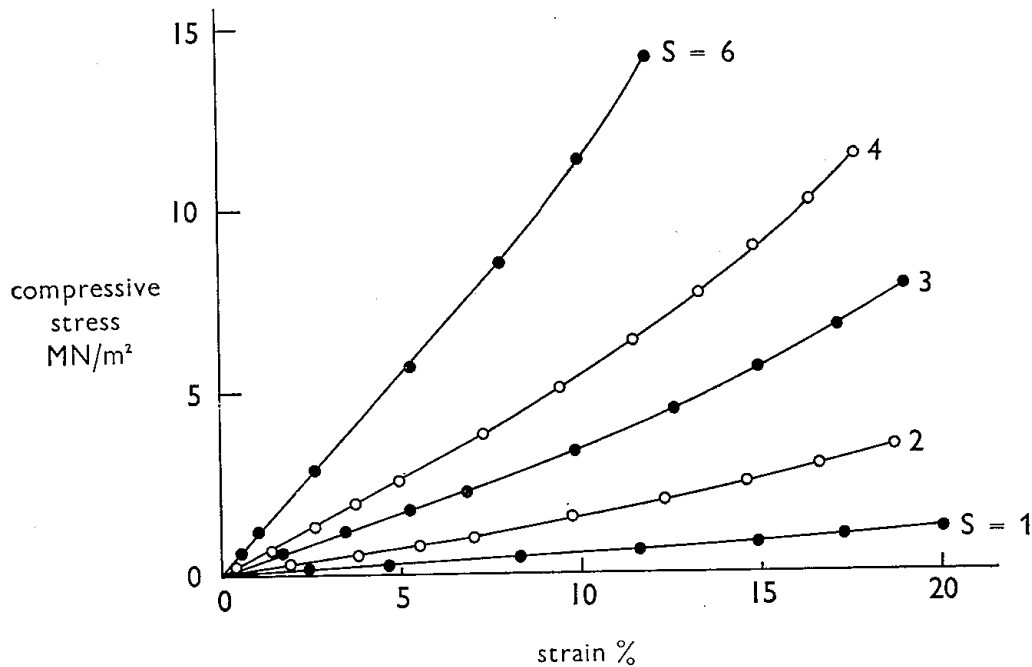


Fig. 27.—Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.

Fig. 1 - Shape Factor

Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.

The stiffness of rubber in compression, when the loaded surfaces are prevented from slipping (by bonding or by mechanical location), depends upon the shape factor S (see Fig. 1), defined as the ratio of one loaded area to the total force-free area, as shown in Fig. 2.



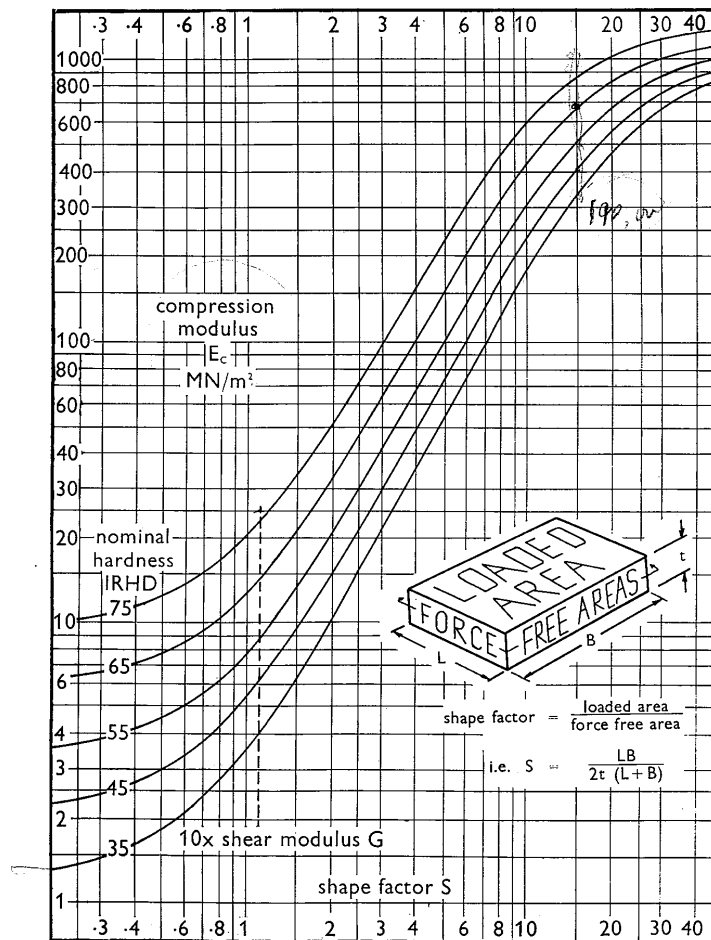


Fig. 28.—Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above).

* Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in E_c when the ratio E_c/E_∞ exceeds about 10%. (E_∞ is the modulus of bulk compression, from Table 3.) To allow for this reduction, use a modified compression modulus obtained by dividing E_c by $1 + (E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred (see footnote to Table 2) then the E_∞ value in Table 3 should be doubled.

Fig. 2 - Compression Modulus

Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above). Chart shows compression modulus E_c (MN/m²) on the vertical axis (log scale, 1 to 1000) versus shape factor S on the horizontal axis (log scale, 0.3 to 40), with separate curves for nominal hardness values of 35, 45, 55, 65, and 75 IRHD, and a line labeled "10x shear modulus G ". An inset diagram illustrates a rubber block showing the "loaded area" (top and bottom faces) versus "force free areas" (side faces), with shape factor = loaded area / force free area, i.e. $S = (LB) / (2t(L+B))$.

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E_c by $1 + (E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred (see footnote to Table 2) then the E_∞ value in Table 3 should be doubled.

0.2 - 2i | Modulus

$$S = (LB) / (2t*(L+B))$$

S = shape factor t = thickness L = length B = breadth

For a block of square section (i.e. $L = B$) or circular section (diameter = L):

$$S = L / (4*t)$$

The compression modulus E_c depends upon the shape factor S (for derivation see Appendix).

$$E_c = E_0(1 + 2kS^2)$$

E_c = compression modulus E_0 = Young's modulus (from Table 3) S = shape factor k = a numerical factor (from Table 3)

See footnote on following page.

When $S > 3$ it may be more convenient to use

$$E_c \approx 5GS^2$$

where G is the shear modulus (from Table 3).

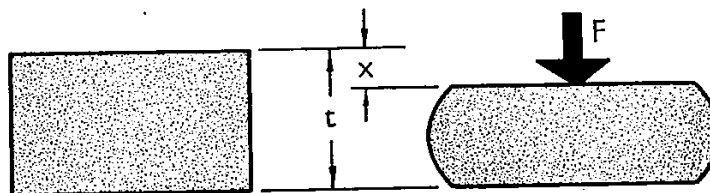


Fig. 29.—Rubber block in compression.

Fig. 3 - Compression Stiffness K_c

Rubber block in compression. Diagram shows a rectangular rubber block of thickness t , deflecting by x under a compressive load F applied to a cylindrical rubber sample.

The compression stiffness, K_c , is given by

$$K_c = F/x = E_c*A/t$$

where F = compressive load E_c = compression modulus (corrected, if necessary, for the effect of bulk compression) A = cross-sectional area t = thickness x = deflexion

It is not advisable to rely on friction alone to prevent slip under a compressive load when using unbonded rubber parts, because slip may occur if $S > \mu/2$, where S is the shape factor and μ the coefficient of friction.



The load (F)—deflexion (x) curve of rubber in compression is non-linear. With no slip it has the approximate form

$$F = E_c A_e (1 + e)$$

where e, the compressive strain, equals x/t . The non-linearity is usually ignored for strains up to about 10%.

There is as yet no method of calculating that a block will be stable but experience has shown that provided the thickness is less than one-quarter of the least plan dimension there should be no instability.

0.2 - 3v | Strip Analysis

COMPRESSION STRIP

When a long strip of rubber is compressed the strain in the direction of its length will be negligible.

Shape factor

$$S = b / (2t)$$

IMAGE | [img/fig30.png](#) | Compression Strip, 80, num, not

Diagram shows a long rectangular strip of rubber of width b, thickness t, and unit length, compressed by a load F applied per unit length.

Compression modulus E_c

$$E_c = 4E_0(1 + kS^2) / 3$$

The compression stiffness per unit length, K_c , is given by

$$K_c = F/x = E_c b/t = (4bE_0(1 + kS^2)) / (3t)$$

where F = load per unit length E_c = compression modulus (corrected, if necessary, for the effect of bulk compression) E_0 = Young's modulus (from Table 3) b = width of strip t = thickness of strip x = deflexion k = a numerical factor (from Table 3) S = shape factor

The load per unit length (F)—deflexion (x) curve for a compressed strip is non-linear. It has the approximate form

$$F = E_c b e (1 + 3e/2)$$

where e, the compressive strain, equals x/t . As in the case of blocks, the non-linearity is usually ignored for strains up to about 10%.



- 1 P.B. Lindley, Engineering Design with Natural Rubber, NR Technical Bulletin. Malaysian Rubber Producers' Research Association, Brickendonbury, U.K.

